AnalogRetriever: Learning Cross-Modal Representations for Analog Circuit Retrieval

Yihan Wang*
Tsinghua University
Beijing, China
yihan-wa23@mails.tsinghua.edu.cn

Lei Li*
The University of Hong Kong
Hong Kong, China
nlp.lilei@gmail.com

Yao Lai
University of Cambridge
Cambridge, United Kingdom
yl2204@cam.ac.uk

Jing Wang†
Nanjing University of Posts and Telecommunications
Nanjing, China
wangjing25@njupt.edu.cn

Yan Lu†
Tsinghua University
Beijing, China
yanlu@tsinghua.edu.cn

Abstract

Analog circuit design relies heavily on reusing existing intellectual property (IP), yet searching across heterogeneous representations such as SPICE netlists, schematics, and functional descriptions remains challenging. Existing methods are largely limited to exact matching within a single modality, failing to capture cross-modal semantic relationships. To bridge this gap, we present AnalogRetriever, a unified tri-modal retrieval framework for analog circuit search. We first build a high-quality dataset on top of Masala-CHAI through a two-stage repair pipeline that raises the netlist compile rate from 22.0% to 99.97%. Built on this foundation, AnalogRetriever encodes schematics and descriptions with a vision-language model and netlists with a port-aware relational graph convolutional network, mapping all three modalities into a shared embedding space via curriculum contrastive learning. Experiments show that AnalogRetriever achieves an average Recall@1 of 75.2% across all six cross-modal retrieval directions, significantly outperforming existing baselines. When integrated into the AnalogCoder agentic framework as a retrieval-augmented generation module, it consistently improves functional pass rates and enables previously unsolved tasks to be completed. Code and data: https://github.com/yihan-wa/AnalogRetriever_ICCAD.

*Both authors contributed equally to this research.
†Corresponding authors.



This work is licensed under a Creative Commons Attribution 4.0 International License.

ICCAD '26, San Jose, CA, USA
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2873-0/2026/11
https://doi.org/10.1145/3831252.3834253

CCS Concepts

• **Hardware** → **Electronic design automation**; • **Information systems** → *Multimedia and multimodal retrieval*; • **Computing methodologies** → *Neural networks.*

Keywords

Analog circuit retrieval; retrieval-augmented generation; cross-modal representation learning; SPICE netlist; graph neural network

ACM Reference Format:
Yihan Wang, Lei Li, Yao Lai, Jing Wang, and Yan Lu. 2026. AnalogRetriever: Learning Cross-Modal Representations for Analog Circuit Retrieval. In *IEEE/ACM International Conference on Computer-Aided Design (ICCAD '26), November 08–12, 2026, San Jose, CA, USA.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3831252.3834253

1 Introduction

Recent advances in large language models (LLMs) have opened new opportunities for analog circuit design automation. Existing efforts focus on *generative* approaches that synthesize designs from specifications or generate netlists from schematic images, but they suffer from hallucination, invalid topologies, and difficulty incorporating domain-specific constraints.

In contrast, *retrieval-based* approaches for analog design remain largely unexplored despite their practical potential. As shown in Fig. 1(a), junior engineers today spend considerable time manually searching design manuals, papers, and internal repositories with keyword queries, a process that is time-consuming, expertise-heavy, and especially hard for newcomers who may not know the right terminology. The challenge is compounded by the heterogeneous nature of the circuit representations: the same circuit exists as a SPICE

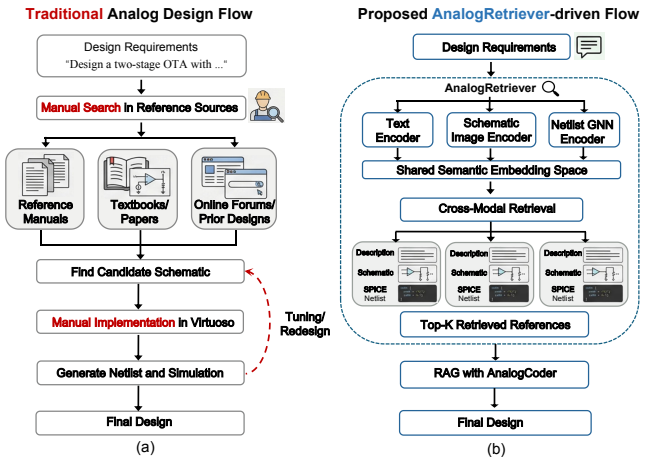


Figure 1. Comparison of analog design flows: (a) Traditional flow based on manual search. (b) Proposed AnalogRetriever-driven flow based on cross-modal retrieval and retrieval-augmented generation.

netlist (code), a schematic (image), or a functional description (text), yet existing circuit retrieval systems operate on a single modality [15].

Unlike digital design, which benefits from mature synthesis and automation flows, analog design relies heavily on reusing proven topologies accumulated through years of engineering experience [4]. An effective cross-modal retrieval system would let engineers describe requirements in natural language and obtain matching schematics and netlists (Fig. 1(b)), accelerating design exploration, facilitating intellectual property (IP) reuse, and lowering the entry barrier for junior designers. Such a system also complements generative methods through retrieval-augmented generation (RAG), where verified existing designs ground LLM outputs and mitigate hallucination. However, building it requires bridging the representational gap between netlists, schematics, and functional descriptions, a cross-modal alignment challenge that remains open.

To address this gap, we propose **AnalogRetriever**, a tri-modal retrieval framework that maps functional descriptions, schematics, and SPICE netlists into a unified semantic space via contrastive learning [20]. This enables flexible cross-modal retrieval: a natural-language query such as "a two-stage op-amp with Miller compensation" returns matching schematics and netlists, and a schematic or netlist query returns functionally similar designs with their specifications.

Building this framework requires addressing two key challenges that no existing method tackles jointly.

(Challenge 1) Domain gap for vision-language alignment. Pretrained vision-language models (VLMs) such as CLIP [16] excel at aligning natural photographs with captions but do not generalize to abstract circuit schematics: their zero-shot average Recall@1 (R@1) is near random (2.5%, Table 2). Circuit schematics are clean line drawings with domain-specific symbols outside CLIP's pretraining distribution, and the largest public dataset (Masala-CHAI) suffers from severe quality issues: only 22% of netlists compile, hindering effective fine-tuning.

(Challenge 2) Graph-structured netlists and fine-grained discrimination. SPICE netlists are fundamentally graph-structured: node names are arbitrary, topology is implicit in connectivity, and standard text encoders cannot capture these structural semantics. Moreover, circuits in the same functional category (e.g., different op-amp topologies) share similar descriptions but differ substantially in implementation, making it easy for a contrastive model to conflate them.

To tackle Challenge 1, we refine Masala-CHAI through a two-stage LLM-based repair pipeline and freeze the lower CLIP layers for domain adaptation without catastrophic forgetting. To tackle Challenge 2, we employ a port-aware Relational Graph Convolutional Network (RGCN) encoder with curriculum-guided hard-negative mining.

We evaluate AnalogRetriever on a curated Masala-CHAI [1], where our two-stage repair pipeline yields 6,371 refined triplets with a 99.97% compilation rate. AnalogRetriever achieves an average R@1 of 75.2% across all six cross-modal directions, outperforming the strongest zero-shot baseline (CROP [15], 4.7%) by over 15×, and the tri-modal training also improves the bi-modal image–text directions. When integrated with AnalogCoder via RAG, it lifts functional correctness on *all eight* evaluated LLMs (averaging +5.6%) and sets a new state of the art of 86.7% on Claude Sonnet 4.6.

Our main contributions are: **(1) Tri-Modal Topology-Aware Retrieval.** We propose AnalogRetriever, which combines a pretrained VLM with a topology-aware graph neural network (GNN) encoder through tri-modal contrastive learning to enable the first cross-modal search across text, schematics, and SPICE netlists. **(2) Curriculum-Guided Hard Negative Mining.** A training strategy that progressively increases intra-cluster negatives based on functional category, sharpening discrimination among structurally distinct circuits with similar functionality. **(3) High-Quality Tri-Modal Dataset.** A two-stage LLM-based refinement pipeline that audits and repairs Masala-CHAI, producing 6,371 simulator-checked triplets that we will release publicly.

2 Related Work

Prior work relevant to AnalogRetriever spans three areas: circuit dataset construction and schematic-netlist conversion, LLM-based analog circuit design, and cross-modal retrieval with contrastive learning. No existing work addresses all three modalities within a single retrieval framework.

2.1 Schematic-Netlist Conversion

AMSNet [19] and AMSNet 2.0 [18] pioneered automatic schematic-to-netlist conversion. Masala-CHAI [1] scaled dataset creation via end-to-end SPICE generation from schematics, and Image2Net [24] contributed more diverse pairs. Wang et al. [23] showed that topology-aware encoders capture circuit semantics beyond structural similarity, and Netlistify [10] tackled deterministic schematic-to-netlist conversion through component recognition. Our framework extends these topology-aware representations to align netlists with schematics and text via contrastive learning.

2.2 LLMs for Analog Circuit Design

AnalogCoder [12] introduced the first training-free LLM agent for analog design via Python code generation; AnalogCoder-Pro [13] extended it with multimodal topology synthesis, and LLMs have likewise been applied to register-transfer level (RTL) code optimization under IP constraints [22]. AnalogXpert [25] formulates topology synthesis as subcircuit-level SPICE generation, Chen et al. [6] curate a textbook-derived corpus to train domain-specific LLMs, LaMAGIC [3] casts topology generation as language modelling, Artisan [5] automates op-amp design end-to-end, and AnalogGenie [9] explicitly explores the topology space. These methods focus on circuit *generation* but suffer from hallucination and invalid topologies. Our retrieval component grounds LLM-based tools with relevant circuit examples via RAG, reducing invalid outputs.

2.3 Cross-Modal Retrieval and Contrastive Learning

Contrastive vision-language pretraining [16, 20] has become the de facto approach for aligning heterogeneous modalities, but its direct application to circuit data is fundamentally limited: SPICE netlists are graph-structured with arbitrary node names, and circuits sharing similar descriptions can differ substantially at the device level. GNNs for electronic design automation (EDA) [11, 17, 23] recover structural semantics from netlists but are typically trained in isolation, without alignment to natural language or schematics. To our knowledge, no prior work establishes a single representation space that bridges all three analog circuit modalities; AnalogRetriever closes this gap and supports all six cross-modal directions with a unified training objective.

3 Method

Fig. 2 overviews the AnalogRetriever framework. We formalize the problem, describe the modality-specific encoders, present the tri-modal contrastive objective, and detail the three-phase curriculum training.

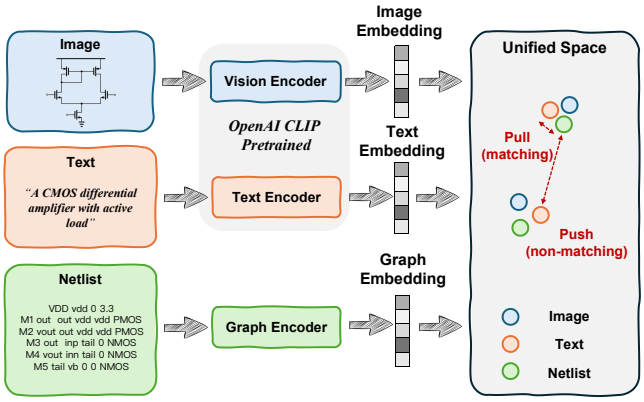


Figure 2. Overview of AnalogRetriever: three modality-specific encoders map images, text, and netlists into a unified embedding space.

3.1 Problem Formulation

We formulate analog circuit retrieval as a tri-modal matching problem across SPICE netlist code $\mathcal{C}$, functional text descriptions $\mathcal{T}$, and schematic images $\mathcal{I}$. Given a dataset $\mathcal{D} = \{(c_i, t_i, s_i)\}_{i=1}^{N}$ of N aligned triplets (code, text, schematic image), we learn three encoders f_C, f_T, f_I that map each modality into a shared d-dimensional embedding space $\mathbb{R}^d$ where semantically related items cluster together. At inference, a query from any modality retrieves top-K items from any target modality via cosine similarity, covering all six cross-modal directions (C↔I, C↔T, I↔T).

3.2 Tri-Modal Encoding Architecture

Vision-Language Encoding with CLIP. We encode schematic images and textual descriptions with the pretrained CLIP [16] VLM: a Vision Transformer (ViT)-L/14 [7] image encoder f_I and a Transformer [21] text encoder f_T, both projecting into a shared d=768 space. To preserve the pretrained cross-modal alignment while allowing domain adaptation to clean circuit line drawings, we freeze the bottom 16 of 24 ViT blocks and fine-tune only the top 8, avoiding catastrophic forgetting of CLIP's general visual-semantic prior.

Port-Aware Relational Graph Netlist Encoding. SPICE netlists are graph-structured. We parse each netlist into a graph whose nodes are both devices and electrical nets: a device is linked to every net it touches by an edge typed with the corresponding terminal, and devices sharing a net are additionally linked by a *shared_net* edge. The net nodes make connectivity explicit, while the *shared_net* edges expose device-level adjacency directly, so L message-passing layers reach L device hops. Crucially, different terminals on the same device carry distinct electrical semantics: connecting to the gate versus the drain of a MOSFET implies a control input rather than a current path. Standard

Graph Convolutional Networks (GCNs) [11] with a single shared edge weight cannot distinguish these port-level differences, confusing functionally distinct circuits that share the same topology. We therefore adopt an RGCN [17] with a separate learnable weight matrix per edge type. As shown in Sec. 4.3, replacing GCN with RGCN brings the largest gains on code-involved retrieval directions, precisely where port-level semantics matter most.

Each netlist c_i is parsed into a graph $G_i=(\mathcal{V}_i, \mathcal{E}_i, \mathcal{R})$ with $|\mathcal{R}|=20$ edge types: 19 port types covering all device terminals—MOSFET (drain/gate/source/bulk), BJT (collector/base/emitter), source (±), diode (anode/cathode), passive R/C/L terminals, four controlled-source ports, *shared_net* for device–device connections on the same electrical net, and *subckt_terminal* for subcircuit interfaces—plus one reserved type for unrecognised ports. This port-level vocabulary lets the encoder distinguish, e.g., a MOSFET whose drain feeds the next stage from one whose source does, which is essential for telling a common-source stage apart from a source follower. Each device node's feature fuses a discrete component-type embedding with a log-normalized continuous parameter vector (channel width and length, resistance, capacitance); net nodes use a net-type embedding (generic, supply, or ground) with a zero parameter vector:

$$\mathbf{h}_v^{(0)} = \mathbf{W}_{\text{fuse}}\big[\text{Emb}(x_{\text{type}}) \,\|\, \text{Linear}(\log(1+x_{\text{cont}}))\big] \quad (1)$$

where $\mathbf{W}_{\text{fuse}}$ is a learnable fusion matrix, $\text{Emb}(\cdot)$ is a discrete embedding lookup, x_{type} is the component type, x_{cont} is the continuous parameter vector, and $\|$ denotes concatenation.

Port-aware message passing at layer l is

$$\mathbf{h}_v^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{R}}\sum_{u\in\mathcal{N}_r(v)}\frac{1}{|\mathcal{N}_r(v)|}\mathbf{W}_r^{(l)}\mathbf{h}_u^{(l)}\right) + \mathbf{h}_v^{(l)}, \quad (2)$$

where $\mathcal{N}_r(v)$ is the set of neighbors of node v under relation type r, $\mathbf{W}_r^{(l)}$ is the relation-specific weight matrix at layer l, and σ denotes the GELU activation. Each layer further applies GraphNorm [2] and a residual connection to stabilize gradient flow. We use L=2 layers: a two-hop receptive field is sufficient to capture the canonical subcircuit patterns that dominate analog topologies (differential pairs, current mirrors, cascode stacks typically span one or two device hops).

A learnable attention pool then aggregates node embeddings into a graph-level vector, letting the model up-weight functionally critical nodes—input pairs, output stages, bias references, and the nets joining them—over boilerplate:

$$\mathbf{g}_i = \sum_{v\in\mathcal{V}_i}\alpha_v\mathbf{h}_v^{(L)}, \quad \alpha_v \propto \exp(\mathbf{w}^\top\mathbf{h}_v^{(L)}), \quad (3)$$

where $\mathbf{w}$ is a learnable attention vector. Compared with uniform sum or mean pooling, attention pooling gives the graph embedding a clearer notion of *which parts of the circuit matter* for a given functional role. Finally, a two-layer multi-layer perceptron (MLP) projection head with LayerNorm, GELU activation, and dropout (p=0.1) maps $\mathbf{g}_i$ into the CLIP space, $\mathbf{v}_i^{(c)}=f_C(c_i)=\text{MLP}_{\text{proj}}(\mathbf{g}_i)$, followed by ℓ_2 normalization. The nonlinear projection provides extra capacity to bridge the representation gap between the graph and vision-language domains without perturbing the pretrained CLIP space.

3.3 Tri-Modal Contrastive Learning

We align the three modalities with an InfoNCE-style [20] objective. Let $\mathbf{v}_i^{(c)}, \mathbf{v}_i^{(s)}, \mathbf{v}_i^{(t)}$ denote the ℓ_2-normalized embeddings of the i-th code, schematic image, and text sample, respectively. For a batch of B triplets, the directional Code→Image loss is

$$\mathcal{L}_{C\to\mathcal{I}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp(\text{sim}(\mathbf{v}_i^{(c)}, \mathbf{v}_i^{(s)})/\tau)}{\sum_{j=1}^{B}\exp(\text{sim}(\mathbf{v}_i^{(c)}, \mathbf{v}_j^{(s)})/\tau)} \quad (4)$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity and τ is a learnable temperature. The full tri-modal loss sums all six directions across the three modality pairs:

$$\mathcal{L}_{\text{tri}} = \sum_{(a,b)\in\mathcal{P}}\big(\mathcal{L}_{a\to b} + \mathcal{L}_{b\to a}\big), \quad \mathcal{P}=\{(C,\mathcal{I}),(\mathcal{T},\mathcal{I}),(C,\mathcal{T})\}. \quad (5)$$

We apply label smoothing ϵ=0.1, which we find especially helpful when training with intra-cluster hard negatives. Importantly, the complete tri-modal training procedure produces *mutual enhancement*: the originally bi-modal Image↔Text directions also improve, so aligning the three modalities benefits all of them.

3.4 Three-Phase Curriculum Training

Jointly training a randomly initialized RGCN with pretrained CLIP is unstable: low-quality graph embeddings disrupt the pretrained alignment, and hard negatives amplify this instability before cross-modal correspondence is even established. We address this with the three-phase curriculum illustrated in Fig. 3, which progressively increases both the number of trainable parameters (RGCN → RGCN+CLIP) and the sample difficulty (α_0=0.05 → $\alpha_{\max}$=0.3) as training proceeds. The figure also makes explicit which loss is active in each phase: Phase 1 uses only the code-involved losses $\mathcal{L}_{I\leftrightarrow C}+\mathcal{L}_{T\leftrightarrow C}$, while Phases 2 and 3 use the full 6-way tri-modal loss $\mathcal{L}_{\text{tri}}$. Phase 1 and the switch to $\mathcal{L}_{\text{tri}}$ are shared by all tri-modal variants in Table 2; the Phase-2 restart and Phase-3 hard negatives are AnalogRetriever-specific.

Phase 1: Graph Encoder Warm-Up. Only the RGCN is trained; both CLIP encoders are frozen. The alignment loss uses only the code-involved directions, $\mathcal{L}_{I\leftrightarrow C}+\mathcal{L}_{T\leftrightarrow C}$, with α=0 (no intra-cluster negatives; see Phase 3). This lets the graph encoder align to the existing CLIP space without perturbing its pretrained structure.

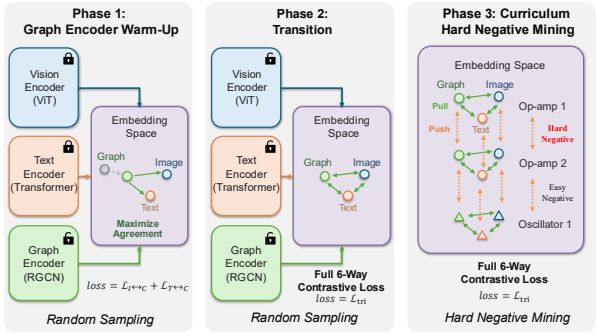


Figure 3. Overview of the three-phase curriculum training strategy.

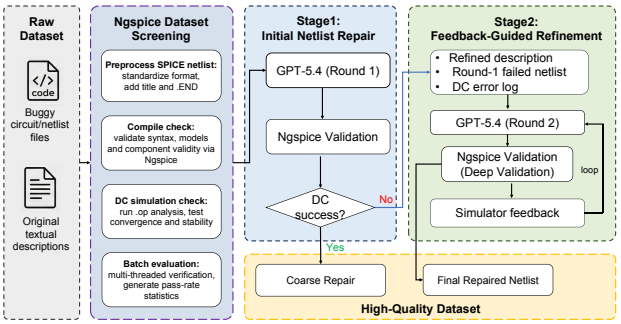


Figure 4. Overview of the two-stage dataset refinement pipeline.

Stage 2a: Netlist Repair

You are an expert SPICE engineer debugging a netlist that failed Ngspice compilation or DC simulation. Given the schematic image, the broken netlist, and the specific Ngspice error messages, return a corrected, compilable netlist that

- resolves all reported syntax and reference errors,
- provides `.model` statements for every semiconductor device,
- preserves the original topology and function,
- includes an `.op` analysis and ends with `.END`.

Output format: `<NETLIST>...</NETLIST>`.

Stage 2b: Description Refinement

You are an expert analog circuit engineer. Using the verified SPICE netlist as ground truth (together with the original, possibly inaccurate caption), produce *one* technically precise sentence (≤60 words, max 77 CLIP tokens) that names the circuit type, lists the key components and their roles, and describes the primary operating principle. Output format: `<DESCRIPTION>...</DESCRIPTION>`.

Figure 5. Stage 2 prompts: feedback-guided netlist repair and description refinement.

Phase 2: Transition. We unfreeze CLIP and switch to the full 6-way loss $\mathcal{L}_{\text{tri}}$ while keeping α=0, so the model is not simultaneously asked to handle newly unfrozen parameters *and* hard negatives. The optimizer is rebuilt with fresh Adam state and an independent linear-warmup-then-cosine-decay schedule, establishing a stable joint-optimisation trajectory before hard negatives are introduced.

Phase 3: Curriculum Hard Negative Mining. We cluster all circuits by embeddings of their SPICE netlists, obtained from Qwen3-Embedding, using K-means (K=30) into functional clusters $\{\mathcal{G}_1, \ldots, \mathcal{G}_K\}$. Each batch is built around a seed circuit drawn uniformly from the training set: an α_m fraction of the batch is drawn from the seed's cluster $\mathcal{G}_k$ and the remainder uniformly from the other clusters. Since the contrastive loss treats every batch item as a positive, the intra-cluster items also act as hard negatives for one another. The ratio increases linearly:

$$\alpha_m = \min\big(\alpha_{\max},\ \alpha_0 + \tfrac{m-1}{M-1}(\alpha_{\max} - \alpha_0)\big), \tag{6}$$

where m is the current training epoch within Phase 3, M is the total number of Phase-3 epochs, α_0=0.05 is the initial hard-negative ratio, and $\alpha_{\max}$=0.3 is the maximum ratio. This schedule first consolidates coarse distinctions (amplifier vs. oscillator) and then progressively sharpens discrimination among structurally distinct circuits serving similar functions, such as common-source vs. common-gate amplifiers or Miller-compensated vs. folded-cascode op-amps.

4 Experiments

We describe dataset curation, training setup and baselines, main retrieval results with ablations, and RAG integration into AnalogCoder.

4.1 Dataset Curation

We build upon Masala-CHAI [1], the largest publicly available tri-modal analog circuit dataset (≈6,500 triplets). Our audit revealed severe quality issues: of the 6,371 schematic images, only 6,069 have paired SPICE netlists and captions, and among those only 22.0% compile under Ngspice and a mere 11.4% pass a DC operating-point (`.op`) check (Table 1). Common failure modes include undefined subcircuits, missing device models, and incorrect node connectivity, and many captions are generic boilerplate (*"This is a circuit"*) that does not reflect the actual topology.

Our refinement pipeline shown in Fig. 4 has two cascaded stages connected by an Ngspice simulator acting as the ground-truth oracle.

Stage 1: Initial Netlist Repair. As shown in the middle block of Fig. 4, each raw netlist is fed to GPT-5.4 [14] together with its Ngspice errors. Repaired netlists that compile *and* pass the DC check are committed to the high-quality set; DC

Table 1. Dataset quality before and after refinement.

Stage	# Triplets	Compile (%)	DC Pass (%)
Original Masala-CHAI	6,069	22.0	11.4
+ Stage 1 (Initial Repair)	6,371	99.2	74.1
+ Stage 2 (Feedback Refinement)	6,371	99.97	99.7
Passing both checks	6,354	100.0	100.0

failures are forwarded to Stage 2 with the error log. For the 302 samples originally missing a netlist or caption, the LLM synthesises both modalities from the schematic image, recovering all 6,371 triplets. Stage 1 alone raises the compile rate from 22.0% to 99.2% and DC pass from 11.4% to 74.1%.

Stage 2: Feedback-Guided Refinement. For Stage-1 DC failures, we use the two prompts in Fig. 5: a *netlist-repair* prompt (Stage 2a) that fixes the broken netlist from the exact Ngspice error messages while preserving topology, and a *description-refinement* prompt (Stage 2b) that rewrites the caption from the now-verified netlist. At each iteration the model receives the failed netlist and DC error log and generates a new candidate verified by Ngspice; updated feedback is appended on failure, for up to $K_{\max}$=5 iterations. Stage 2 lifts the compile rate to 99.97% and the DC pass rate to 99.7% (Table 1), ensuring text–circuit consistency. A before/after example appears in Fig. 6.

4.2 Experimental Settings

Preprocessing and Training. We split all 6,371 triplets at the instance level with a seeded random shuffle (seed 42), hold out the first 1,000 for testing, and train on the remaining 5,371. Functional clusters for the curriculum sampler are obtained by K-means (K=30) on Qwen3-Embedding representations of the SPICE netlists. We use CLIP ViT-L/14 [16] (bottom 16 of 24 blocks frozen), a two-layer RGCN [17] with hidden dim 512 and attention pooling, and a two-layer MLP into the 768-d CLIP space. Training follows the three-phase curriculum of Sec. 3.4 with 6, 2, and 12 epochs per phase; α ramps from 0.05 to 0.3 in Phase 3. We use AdamW (CLIP 5×10^{-5}, RGCN 5×10^{-4}), rebuilt at Phase 2 with an independent warmup, effective batch size 256, and a learnable logit scale ($1/\tau$) initialised to $1/0.07$. Checkpoints are selected by held-out R@1 (mean of I→C, T→I, and T→C; T→I for TI). All experiments run on a single NVIDIA H200 GPU.

Metrics and Baselines. We report Recall@K ($K\in\{1,5,10\}$) for all six cross-modal directions and the average R@1 over them. Each evaluation draws a gallery of N=200 items uniformly from the 1,000 held-out triplets, ranks candidates by cosine similarity, and averages over ten draws with fixed seeds (random-guess R@1 is 0.5%). Exact-pair Recall@K is a strict lower bound on usefulness, since several library circuits can satisfy one query. We compare against four external retrieval baselines, all applied zero-shot without

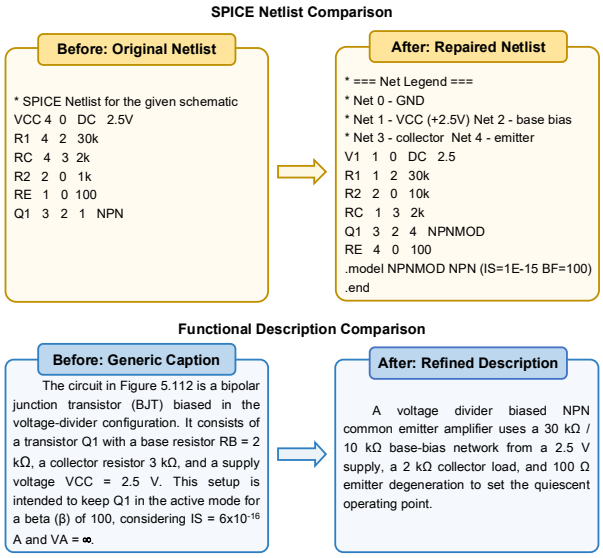


Figure 6. Before/after refinement. Top: SPICE netlist with errors and fixes. Bottom: generic vs. refined functional description.

training on our corpus: (i) **CLIP** [16] (raw SPICE fed to the off-the-shelf CLIP text encoder, a zero-shot lower bound); (ii) **CROP** [15] (netlists summarized by Qwen2.5-7B then CLIP-embedded, a training-free code-as-text baseline); (iii) **ChatLS** [26] (LLM-based structured netlist representations); and (iv) **NetTAG** [8] (per-device text attributes encoded with Qwen3-Embedding and pooled). CROP, ChatLS, and NetTAG only replace the *code* representation, so all four baselines share the off-the-shelf CLIP image–text pathway and yield identical T→I and I→T recall. Following the notation of Sec. 3.1 (I: image, T: text, C: circuit netlist), the internal ablation variants **TI (Bi-Modal)**, **TIC (GCN)**, and **TIC (RGCN)** share the 20-epoch budget: TI fine-tunes CLIP on image–text pairs only, while TIC applies the Phase-1 warm-up followed by the six-way loss but omits the Phase-2 optimizer restart and the Phase-3 hard negatives.

4.3 Main Results and Ablation

We present all results, including external baselines and internal ablations, in Table 2 to facilitate direct comparison.

Tri-modal training benefits bi-modal retrieval. On the shared Image↔Text directions, the full tri-modal procedure lifts T→I R@1 from 70.5% to **78.2%** (+7.7) and I→T from 69.8% to **78.5%** (+8.7) over the bi-modal TI baseline, despite using the same CLIP backbone and identical image–text data. Notably, adding the graph encoder alone (TIC vs. TI) barely moves T2I (70.5→70.8): even with the graph encoder warmed up against frozen CLIP, joint training without hard negatives does not by itself sharpen the image–text space.

Table 2. Cross-modal retrieval on the held-out set (Recall@K, %). Top: zero-shot baselines; bottom: models trained on our corpus. Best per column in bold.

	I→C			T→I			T→C			C→I			I→T			C→T			Avg
Model	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	**R@1**
CLIP [16]	1.2	4.6	8.9	2.2	8.2	13.5	2.9	9.7	16.2	2.9	10.1	15.7	2.7	7.6	12.9	3.2	11.2	17.4	2.5
CROP [15]	2.3	8.4	15.0	2.2	8.2	13.5	9.2	25.8	37.1	2.4	8.5	14.2	2.7	7.6	12.9	9.4	24.7	33.0	4.7
ChatLS [26]	0.9	5.9	11.3	2.2	8.2	13.5	9.5	24.0	35.1	1.6	6.6	10.0	2.7	7.6	12.9	7.2	20.4	31.3	4.0
NetTAG [8]	0.2	2.5	4.5	2.2	8.2	13.5	0.5	3.4	6.8	0.1	1.4	4.1	2.7	7.6	12.9	0.4	3.4	7.1	1.0
TI (Bi-Modal)	–	–	–	70.5	92.9	96.8	–	–	–	–	–	–	69.8	94.1	96.7	–	–	–	–
TIC (GCN)	62.9	92.2	95.7	70.8	95.7	98.4	67.7	95.7	99.0	60.9	91.6	95.6	71.3	96.4	98.8	65.5	95.7	98.9	66.5
TIC (RGCN)	65.2	92.5	96.8	71.8	96.2	**98.8**	69.1	**96.4**	**99.2**	61.9	92.9	96.7	71.1	**96.7**	**98.9**	67.3	96.4	**99.0**	67.7
AnalogRetriever	**74.7**	**95.5**	**98.1**	**78.2**	**96.8**	97.9	**75.6**	96.3	98.6	**72.0**	**94.8**	**97.9**	**78.5**	**96.7**	98.3	**72.4**	**96.7**	98.3	**75.2**

Table 3. Functional correctness (%) on the AnalogCoder benchmark (24 tasks × 5 trials) for eight LLMs without and with AnalogRetriever; Δ: absolute gain.

LLM	Baseline	+RAG	Δ
GPT-4o-mini	30.8	40.8	**+10.0**
Nemotron-120B	34.2	37.5	+3.3
GPT-5.4-mini	65.0	67.5	+2.5
Gemini-3-Flash	65.0	70.0	+5.0
Kimi-K2	65.0	73.3	+8.3
GLM-4.6	73.3	80.0	+6.7
Qwen3.5-397B	78.3	85.0	+6.7
Claude-Sonnet-4.6	84.2	**86.7**	+2.5
Average	**62.0**	**67.6**	**+5.6**

The bi-modal gains appear only once the Phase-2 restart and the Phase-3 hard negatives are added.

Curriculum hard-negative mining yields consistent gains. Since TIC (RGCN) already includes the Phase-1 warm-up and the six-way loss, comparing it with the full AnalogRetriever isolates the Phase-2 restart and the Phase-3 hard-negative curriculum, which together yield uniform improvements across all six directions (+9.5 on I→C, +10.1 on C→I, +6.4 on T→I, +7.4 on I→T, +6.5 on T→C, and +5.1 on C→T). The largest gains appear on *code-involved* directions, where the hard-negative sampler forces the model to distinguish circuits that share similar high-level descriptions but differ in transistor-level topology (e.g., a folded-cascode op-amp versus a two-stage Miller op-amp). Overall, average R@1 improves from 67.7% to **75.2%** (+7.5). Port-aware RGCN adds +1.2 Avg R@1 over the edge-agnostic GCN by encoding device-port semantics, with the gains concentrated on the code-involved directions. Every AnalogRetriever direction exceeds 94% at R@5 and 97% at R@10, so a user query almost always places the correct circuit within the top-10 candidates.

4.4 Retrieval-Augmented Generation

To demonstrate practical utility, we integrate AnalogRetriever into a RAG pipeline with AnalogCoder [12], a training-free LLM agent for analog design via PySpice code generation.

RAG Pipeline. AnalogRetriever ranks the training-set netlists by cosine similarity to the encoded query via Text→Code search, and a device-family gate keeps the highest-ranked candidates whose dominant device family matches the family inferred from the task description. The selected netlists are converted to PySpice by a rule-based parser *before* prompt injection, so in-context examples use the same API as the target code.

Cross-Model Evaluation. We evaluate on AnalogCoder's 24-task benchmark (5 trials × 3 retries) across eight LLMs (Table 3). AnalogRetriever delivers a positive gain on *all eight* models, averaging **+5.6%** absolute (62.0%→67.6%). The largest gain is on GPT-4o-mini (+10.0); augmenting Claude Sonnet 4.6 reaches **86.7%**, a new state of the art. The benefit *generalizes across model families and scales.*

Qualitative Case Studies. Fig. 7 shows two representative cases. In Task 9 (top row), Claude Sonnet 4.6 without RAG produces a diff-pair with a current-mirror load but no common-mode feedback, letting the high-impedance node drift into triode (0/5). Grounded by a retrieved single-ended CMOS Miller op-amp, the LLM reproduces the canonical M1→M3 topology with a Miller capacitor, reaching 5/5. In Task 17 (bottom row), GPT-5.4-mini without retrieval produces an incorrect RC network structure and leaves the feedback loop open, preventing oscillation (0/5). With a retrieved reference, the LLM constructs the correct Wien-bridge topology with a closed feedback loop, achieving 4/5 success. Both cases demonstrate that retrieval provides the topological guidance needed to turn invalid outputs into functional circuits.

Effect of Query Expansion. AnalogCoder task prompts are typically only 3–6 tokens long (e.g., *"A Wien Bridge oscillator"*), which CLIP's text encoder maps to low-discriminative embeddings. Rewriting each task into a topology-aware specification lifts relevant database entries from deep ranks to the top-k (e.g., the canonical Wien-bridge op-amp entries move from ranks 74/56/20 to 2/10/5; the Miller op-amp from

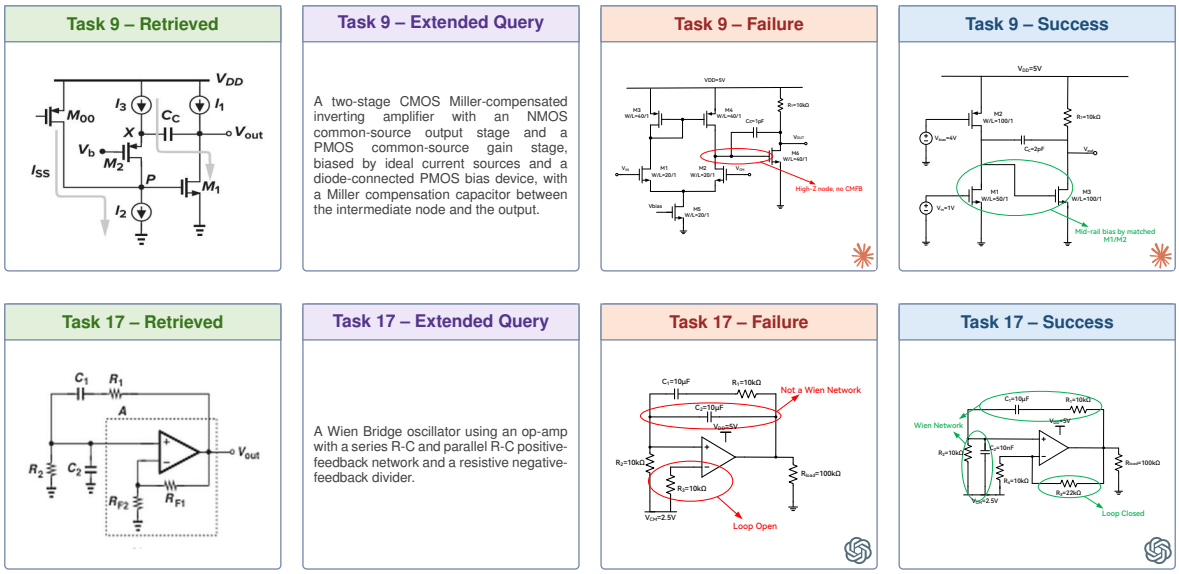


Figure 7. Case studies of retrieval-augmented circuit generation on Tasks 9 and 17.

rank 35 to rank 1). We therefore let GPT-5.4 rewrite every task once into a topology-aware paragraph before retrieval; the cost is amortized across all 24 tasks and negligible. Fig. 7 visualizes both the extended query and its top-1 reference for each case study; all RAG results use expanded queries.

5 Conclusion

We presented AnalogRetriever, the first tri-modal retrieval model that aligns natural-language descriptions, schematic images, and SPICE netlists in a shared embedding space, supported by a two-stage LLM-based refinement pipeline that lifts the Masala-CHAI DC pass rate from 11.4% to 99.7% and yields 6,371 refined triplets. A port-aware RGCN encoder trained with a three-phase curriculum and hard-negative mining pushes average R@1 to **75.2%** across all six cross-modal directions, outperforming the best zero-shot prior baseline by over an order of magnitude. Plugged into AnalogCoder, AnalogRetriever delivers a positive functional-correctness gain on *all eight* evaluated LLMs (averaging +5.6% absolute) and sets a new state of the art of **86.7%** on Claude Sonnet 4.6, with the benefit generalizing across model families and parameter scales rather than being tied to any specific LLM.

Three findings from our experiments deserve particular emphasis. First, *tri-modal training yields mutual enhancement*: even the Image↔Text directions, which share the same CLIP backbone and identical training data, improve by up to +8.7 R@1 over the bi-modal baseline under the full tri-modal procedure, showing that the graph encoder and the hard-negative curriculum together enrich the shared embedding space in ways that benefit all modalities, not just the newly introduced one. Second, *the hard-negative curriculum, not the graph encoder alone, drives the gain*: a tri-modal model that already warms up the graph encoder and trains the full six-way loss still trails AnalogRetriever by +7.5 Avg R@1, which the Phase-2 optimizer restart and the Phase-3 curriculum recover. Third, our qualitative case studies show that a single high-quality retrieved reference is often enough to re-anchor a generation that would otherwise commit to a topologically invalid circuit family, turning complete failures into reliable successes, suggesting that retrieval and generation are complementary rather than competing paradigms for analog design automation.

Limitations and future work. Several directions remain open. (i) The current dataset covers 19 canonical analog topologies; extending to mixed-signal, RF, and power-management circuits would broaden applicability. (ii) The RGCN uses 19 hand-defined port types; learning the relation vocabulary from data could improve generalization to novel device technologies. (iii) As circuit databases grow to industrial scale, efficient nearest-neighbor search (e.g., product quantization) becomes essential. We will release the curated dataset and model weights.

Acknowledgments

This work was supported by the National Key Laboratory of Integrated Circuits and Microsystems (No. NICL2026KF1004) and the Natural Science Research Start-up Foundation of Recruiting Talents of Nanjing University of Posts and Telecommunications (Grant No. NY225066).

References

[1] Jitendra Bhandari, Vineet Bhat, Yuheng He, Hamed Rahmani, Siddharth Garg, and Ramesh Karri. 2025. Masala-CHAI: A Large-Scale SPICE Netlist Dataset for Analog Circuits by Harnessing AI. arXiv:2411.14299

[2] Tianle Cai, Shengjie Luo, Keyulu Xu, Di He, Tie-Yan Liu, and Liwei Wang. 2021. GraphNorm: A Principled Approach to Accelerating Graph Neural Network Training. In *Proceedings of the 38th International Conference on Machine Learning*, Vol. 139. PMLR, 1204–1215. https://proceedings.mlr.press/v139/cai21e.html

[3] Chen-Chia Chang, Yikang Shen, Shaoze Fan, Jing Li, Shun Zhang, Ningyuan Cao, Yiran Chen, and Xin Zhang. 2024. LaMAGIC: Language-Model-based Topology Generation for Analog Integrated Circuits. In *Proceedings of the 41st International Conference on Machine Learning*, Vol. 235. PMLR, 6253–6262. https://proceedings.mlr.press/v235/chang24c.html

[4] Lei Chen, Yiqi Chen, Zhufei Chu, et al. 2024. Large circuit models: opportunities and challenges. *Science China Information Sciences* 67, 10 (2024). doi:10.1007/s11432-024-4155-7

[5] Zihao Chen, Jiangli Huang, Yiting Liu, Fan Yang, Li Shang, Dian Zhou, and Xuan Zeng. 2024. Artisan: Automated Operational Amplifier Design via Domain-specific Large Language Model. In *Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC '24)*. Article 39, 6 pages. doi:10.1145/3649329.3655903

[6] Zihao Chen, Ji Zhuang, Jinyi Shen, Xiaoyue Ke, Xinyi Yang, Mingjie Zhou, Zhuoyao Du, Xu Yan, Zhouyang Wu, Zhenyu Xu, Jiangli Huang, Li Shang, Xuan Zeng, and Fan Yang. 2026. Dataset Construction for Training LLM to Learn Analog Circuit Knowledge. arXiv:2508.10409

[7] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations*. https://openreview.net/forum?id=YicbFdNTTy

[8] Wenji Fang, Wenkai Li, Shang Liu, Yao Lu, Hongce Zhang, and Zhiyao Xie. 2025. NetTAG: A Multimodal RTL-and-Layout-Aligned Netlist Foundation Model via Text-Attributed Graph. In *Proceedings of the 62nd Annual ACM/IEEE Design Automation Conference (DAC '25)*. IEEE Press, Article 430, 7 pages. doi:10.1109/DAC63849.2025.11133349

[9] Jian Gao, Weidong Cao, Junyi Yang, and Xuan Zhang. 2025. AnalogGenie: A Generative Engine for Automatic Discovery of Analog Circuit Topologies. In *International Conference on Learning Representations*, Vol. 2025. 96496–96514.

[10] Chun-Yen Huang, Hsuan-I Chen, Hao-Wen Ho, Pei-Hsin Kang, Mark Po-Hung Lin, Wen-Hao Liu, and Haoxing Ren. 2025. Netlistify: Transforming Circuit Schematics into Netlists with Deep Learning. In *2025 ACM/IEEE 7th Symposium on Machine Learning for CAD (MLCAD)*. IEEE Press, 1–8. doi:10.1109/MLCAD65511.2025.11189145

[11] Thomas N. Kipf and Max Welling. 2017. Semi-Supervised Classification with Graph Convolutional Networks. In *International Conference on Learning Representations*. https://openreview.net/forum?id=SJU4ayYgl

[12] Yao Lai, Sungyoung Lee, Guojin Chen, Souradip Poddar, Mengkang Hu, David Z. Pan, and Ping Luo. 2025. AnalogCoder: Analog Circuit Design via Training-Free Code Generation. *Proceedings of the AAAI Conference on Artificial Intelligence* 39, 1 (2025), 379–387. doi:10.1609/aaai.v39i1.32016

[13] Yao Lai, Souradip Poddar, Sungyoung Lee, Guojin Chen, Mengkang Hu, Bei Yu, Ping Luo, and David Z. Pan. 2026. AnalogCoder-Pro: Unifying Analog Circuit Generation and Optimization via Multi-modal LLMs. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* (2026), 1–1. doi:10.1109/TCAD.2026.3673493

[14] OpenAI. 2026. Introducing GPT-5.4. https://openai.com/index/introducing-gpt-5-4/ Accessed: 2026-04-05.

[15] Jingyu Pan, Isaac Jacobson, Zheng Zhao, Tung-Chieh Chen, Guanglei Zhou, Chen-Chia Chang, Vineet Rashingkar, and Yiran Chen. 2025. CROP: Circuit Retrieval and Optimization with Parameter Guidance using LLMs. In *2025 IEEE/ACM International Conference On Computer Aided Design (ICCAD)*. 1–9. doi:10.1109/ICCAD66269.2025.11240643

[16] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning*, Vol. 139. PMLR, 8748–8763. https://proceedings.mlr.press/v139/radford21a.html

[17] Michael Schlichtkrull, Thomas N. Kipf, Peter Bloem, Rianne van den Berg, Ivan Titov, and Max Welling. 2018. Modeling Relational Data with Graph Convolutional Networks. In *The Semantic Web*. Springer International Publishing, 593–607.

[18] Yichen Shi, Zhuofu Tao, Yuhao Gao, Li Huang, Hongyang Wang, Zhiping Yu, Ting-Jung Lin, and Lei He. 2025. AMSnet 2.0: A Large AMS Database with AI Segmentation for Net Detection. In *2025 IEEE International Conference on LLM-Aided Design (ICLAD)*. 242–248. doi:10.1109/ICLAD65226.2025.00014

[19] Zhuofu Tao, Yichen Shi, Yiru Huo, Rui Ye, Zonghang Li, Li Huang, Chen Wu, Na Bai, Zhiping Yu, Ting-Jung Lin, and Lei He. 2024. AMSNet: Netlist Dataset for AMS Circuits. In *2024 IEEE LLM Aided Design Workshop (LAD)*. 1–5. doi:10.1109/LAD62341.2024.10691781

[20] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2019. Representation Learning with Contrastive Predictive Coding. arXiv:1807.03748

[21] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is All you Need. In *Advances in Neural Information Processing Systems*, Vol. 30. Curran Associates, Inc.

[22] Jing Wang, Zheng Li, Lei Li, Fan He, Liyu Lin, Yao Lai, Yan Li, Xiaoyang Zeng, and Yufeng Guo. 2025. Principle-Guided Verilog Optimization: IP-Safe Knowledge Transfer via Local-Cloud Collaboration. arXiv:2508.05675

[23] Ziyi Wang, Chen Bai, Zhuolun He, Guangliang Zhang, Qiang Xu, Tsung-Yi Ho, Bei Yu, and Yu Huang. 2022. Functionality matters in netlist representation learning. In *Proceedings of the 59th ACM/IEEE Design Automation Conference (DAC '22)*. 61–66. doi:10.1145/3489517.3530410

[24] Haohang Xu, Chengjie Liu, Qihang Wang, Wenhao Huang, Yongjian Xu, Weiyu Chen, Anlan Peng, Zhijun Li, Bo Li, Lei Qi, Jun Yang, Yuan Du, and Li Du. 2025. Image2Net: Datasets, Benchmark and Hybrid Framework to Convert Analog Circuit Diagrams into Netlists. In *2025 International Symposium of Electronics Design Automation (ISEDA)*. 807–816. doi:10.1109/ISEDA65950.2025.11100581

[25] Haoyi Zhang, Shizhao Sun, Yibo Lin, Runsheng Wang, and Jiang Bian. 2025. AnalogXpert: Automating Analog Topology Synthesis by Incorporating Circuit Design Expertise into Large Language Models. In *2025 International Symposium of Electronics Design Automation (ISEDA)*. 772–777. doi:10.1109/ISEDA65950.2025.11100627

[26] Haisheng Zheng, Haoyuan Wu, and Zhuolun He. 2025. ChatLS: Multimodal Retrieval-Augmented Generation and Chain-of-Thought for Logic Synthesis Script Customization. In *2025 62nd ACM/IEEE Design Automation Conference (DAC)*. 1–7. doi:10.1109/DAC63849.2025.11132969